Cairo University

Faculty of Engineering

Computer Engineering Department

Big Data Project - Phase 1

Team 5

Name	BN
Ahmed Hamed Gaber Hamed	3
Akram Hany Karam Salam	13
Amir Anwar Bekhit Awd Kedis	14
Somia Saad Esmail Elshemy	26

Problem Definition

What combination of habits leads to the best academic performance?

Datasets:

College Students Habits & Performance

Planned Approach:

- **Data pre-processing:** Handling missing values, encoding categorical attributes, scaling numerical features, and extracting meaningful features
- **EDA:** We will be using statistical summaries and visualizations such as correlation matrices and distribution plots to understand relationships between student habits and academic performance.
- **Descriptive Analysis:** Clustering techniques such as K-Means and association rules will be applied to segment students into groups (e.g., high performers, average students, and at-risk students) based on lifestyle and academic features.
- **Predictive Analysis:** regression and classification models such as Linear Regression and Logistic Regression will be used to predict GPA or classify students based on performance levels.
- **Model evaluation:** will be performed using metrics such as accuracy, RMSE, precision, and recall.
- **K-Means clustering algorithm will be implemented using MapReduce.**